

Cardiac Field Artifact Removal in Ear-EEG Using Repetitive Disturbance Rejection

Wenlin Zhang¹, Sam Lin^{1,6}, Shuai Li³, Ye Yuan¹,
Zheng Wang^{1,2}, Ziteng Ni⁴, Heng Kang Yao^{2,4}, Yongcheng Li⁵, Davide Momi⁷

Abstract—Continuous wearable EEG is increasingly paired with consumer wearables that provide a heartbeat stream (e.g., smartwatch ECG/PPG). Taking the available ECG/heartbeat reference, we propose a repetitive-control framework that models cardiac artifact as a heartbeat-aligned periodic disturbance. Each EEG sample is assigned a cardiac phase index wrapped around ECG R-peaks; phase-synchronous averaging then preserves heartbeat-correlated residue while attenuating non-phase-locked noise. Subtracting this phase-indexed average yields effective artifact removal in the non-causal offline setting. For online real-time operation, an iterative-learning estimator is used to track the residue in either a same-beat predictive mode (strict causal) or a one-beat-delayed mode (buffered causal). We also propose a PI based phase tracker for robust heartbeat timing estimation from raw EKG waveforms to account for breathe related heart rate variability or immature beats rejection. Results on open datasets and semi-synthetic simulations show that the heartbeat-timing-anchored repetitive/iterative learning approach provides strong artifact suppression at low compute cost, making it well suited to always-on wearable execution, where blind source separation is typically channel-starved.

I. INTRODUCTION

Wearable EEG is moving toward continuous, day-scale monitoring, and many users already wear a second device that provides heartbeat timing from ECG or PPG. This creates a practical path for low-channel ear-EEG denoising: use heartbeat timing already available from the wearable ecosystem rather than requiring additional EEG channels. In ear-centered montages, cardiac field artifact (CFA) is often prominent because electrodes are close to neck and vascular pathways. CFA is the volume-conducted electrical field of the heart [1], distinct from the scanner-related ballistocardiogram artifact emphasized in EEG-fMRI work [2]. For out-of-scanner earables, CFA is therefore a primary contamination mechanism.

Classical ECG-referenced adaptive cancellation is well established [3], and it is a natural baseline. However, in our target regime it faces two practical limits: earables are channel-starved for blind source separation [6], and reference-to-artifact waveform predictability can be weak under real volume-conduction mismatch and nonstationarity. Our real-data observations in this paper (Fig. 1) support exactly this point: coherence between ECG and ear-EEG artifact is close to

zero even with beat-synchronous pooling, so reference-driven waveform regression is limited as a primary mechanism. At the same time, beat timing remains highly informative, because the artifact is still phase-locked to R-peaks.

This motivates a timing-only control view: model cardiac contamination as a repetitive disturbance and reject it by heartbeat phase indexing. Repetitive control and iterative learning control (ILC) are mature tools for periodic disturbance rejection in mechatronics [8], and they map naturally to the heartbeat cycle when robust R-peak timing is available. We therefore use a min/max envelope tracker with PI/PID phase tracking and arrhythmia-aware supervision to produce stable beat timing, then apply phase-aligned template learning/cancellation in three deployment modes: non-causal offline, buffered causal (one-beat latency), and strict real-time causal (zero added latency).

The resulting contribution is a compact, wearable-oriented CFA removal scheme aligned with realistic device constraints: low channel count, low compute budget, and timing references that may come from a separate wearable. We evaluate the design on open real recordings and semi-synthetic data, compare against linear RLS-based and ICA baselines, and quantify the latency-versus-suppression tradeoff between non-causal, buffered, and strict-causal operation.

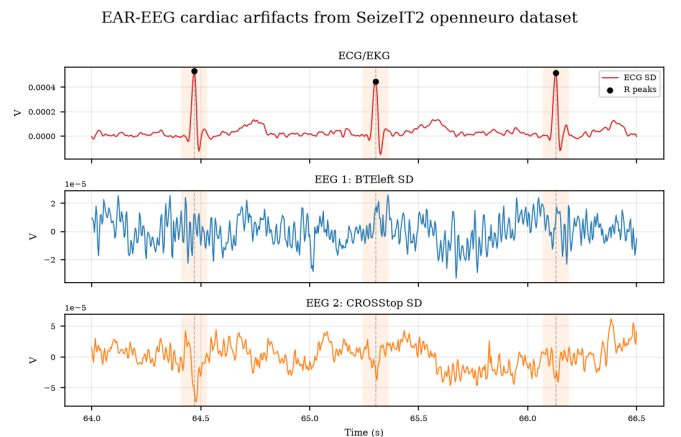


Fig. 1. Short aligned segment from SeizeIT2 (ds005873, sub-001, ses-01, run-01), showing chest ECG/EKG and two ear-EEG channels over about three heartbeats. R-peak neighborhoods are highlighted to emphasize heartbeat-locked contamination timing. For this panel, MNE preprocessing is applied to EEG channels only: 50 Hz notch plus 0.5 Hz high-pass (DC-removal) filtering.

Affiliations: ¹Institute of Brain and Mind, CA, USA; ²Krembil Centre for Neuroinformatics, Centre for Addiction and Mental Health, Toronto, Ontario, Canada; ³University of Oulu, Oulu, Finland; ⁴Synchroni Scientific Instruments Inc., Canada; ⁵Shenzhen Institutes of Advanced Technology, China; ⁶Palos Verdes Peninsula High School, CA, USA; ⁷University of California, Irvine, CA, USA; Corresponding author: Wenlin Zhang (e-mail: william@ibmind.org).

II. PROBLEM FORMULATION

Following repetitive-control modeling conventions [8], [9], we model the measured ear-EEG signal as

$$y(t) = s(t) + d_r(t) + n(t), \quad (1)$$

where $s(t)$ is the neural signal of interest, $d_r(t)$ is the heartbeat-synchronous repetitive cardiac disturbance, and $n(t)$ aggregates asynchronous interference and sensor noise. The subscript r denotes the repetitive nature of the disturbance. Its period is heartbeat dependent and is set by ECG R-peak timing, with beat interval

$$T_k = t_{k+1} - t_k, \quad (2)$$

where t_k is the k -th detected R-peak instant.

Assumption 1. In ear-EEG, CFA is dominated by the repetitive component $d_r(t)$. The R-peak phase-asynchronous disturbances and noise $n(t)$ are assumed zero-mean across the R-peak sync averaging, and DC offset and slow drift are removed by preprocessing.

Assumption 2. The neural signal of interest $s(t)$ is not phase-locked to the R-peak: across beats, its phase relative to the cardiac cycle is effectively uniform. Consequently $s(t)$ is preserved by the proposed phase-synchronous repetitive runoff cancellation, more detailed analysis is provided in Sec. III.

Using heartbeat phase, we represent the repetitive disturbance as a phase-indexed waveform

$$d_r(t) = d_r(\phi(t)), \quad \phi(t) = \frac{t - t_k}{T_k}, \quad t \in [t_k, t_{k+1}), \quad (3)$$

with $\phi(t) \in [0, 1)$ the R-peak-aligned cardiac phase.

Assumption 3. The beat period T_k is assumed slowly varying, dominated by respiratory sinus arrhythmia (RSA), with additional small modulation from low-level muscle/motion effects in resting-state recordings.

Problem 1. Find the repetitive cardiac artifact template $\hat{d}_r(\phi(t))$ such that subtracting it from the measured signal recovers the neural signal of interest:

$$\hat{s}(t) = y(t) - \hat{d}_r(\phi(t)) \approx s(t). \quad (4)$$

Under Assumptions 1–3, $\hat{d}_r(\phi(t))$ can be estimated from the R-peak-locked structure of $y(t)$ alone.

Note 1. Since $d_r(\phi(t))$ is periodic in the cardiac phase $\phi(t)$ indexed solely by R-peak instants $\{t_k\}$, Problem 1 depends only on R-peak timing and never on the ECG waveform shape. Consequently, any wearable heartbeat source—smartwatch ECG, wrist PPG, or any device delivering beat timing via a synchronisation mechanism—is sufficient to instantiate the solution.

Note 2. In reality, $s(t)$ may contain a phase-synchronous part that has non-zero expectation mean value across beats, e.g. the heartbeat-evoked potential (HEP)—a genuine cortical response time-locked to the cardiac cycle may or may not be

absorbed into $\hat{d}_r$. In this paper, we make no claim that the solution of Eq. 4 preserves the phase-synchronous part of the neural signals, and it's a meaningful future work or extension worthy of further investigation.

III. REPETITIVE DISTURBANCE REJECTION

To instantiate Problem 1 on the discrete sample grid, let $\hat{t}_k$ be the estimated k -th R-peak sample instant and f_s the sampling rate. The k -th beat's R-peak-aligned $d_r(\phi(t))$ at integer offset p is defined as

$$d_k\left(\hat{t}_k + \frac{p}{f_s}\right) \triangleq d_k(\phi_p), \quad (5)$$

with the discrete phase index and offset range

$$\phi_p \triangleq \phi\left(\hat{t}_k + \frac{p}{f_s}\right) = \frac{p}{f_s T_k}, \quad (6a)$$

$$p \in \{-n_{\text{pre}}, \dots, -1, 0, 1, \dots, n_{\text{post}}\}. \quad (6b)$$

Note 3. The template window spans $n_t = n_{\text{pre}} + n_{\text{post}} + 1$ samples ($p = 0$ at the R-peak). A practical choice is $n_{\text{pre}} = \lfloor 0.05 f_s \rfloor$ (50 ms, covering QRS onset) and $n_{\text{post}} = \lfloor 0.55 f_s \rfloor$ (550 ms, covering the full QRS-T complex); at $f_s = 250$ Hz this gives $n_t = 151$ samples.

A. Problem 1.1: Non-Causal Cancellation

The non-causal solution to Problem 1 uses all N_r beats to form the minimum-MSE template by same-phase averaging across beats,

$$\hat{d}_r(\phi_p) = \frac{1}{N_r} \sum_{k=1}^{N_r} d_k(\phi_p), \quad (7)$$

enabling CFA removal on the sampled grid as

$$\hat{s}\left(\hat{t}_k + \frac{p}{f_s}\right) = d_k(\phi_p) - \hat{d}_r(\phi_p). \quad (8)$$

Under Assumption 1, the phase-asynchronous noise contribution to $\hat{d}_r(\phi_p)$ averages toward zero as N_r grows.

B. Two-Pass Procedure

Pass 1 — Template estimation. Scan the full EEG recording once with the set of N_r detected R-peak instants $\{\hat{t}_k\}_{k=1}^{N_r}$. For each phase index p (Eq. (6a)), collect the EEG sample $y(\hat{t}_k + p/f_s)$ from every beat k and accumulate the sum. After scanning all beats, divide by the per-index count to form the template $\hat{d}_r(\phi_p)$ per Eq. (7). The result is a single waveform of length $n_t = n_{\text{pre}} + n_{\text{post}}$ that captures the mean R-peak-locked cardiac disturbance over the entire session.

Pass 2 — Repetitive disturbance subtraction. Scan the EEG recording a second time. For each sample $y(t)$, look up the template value at the phase index $\hat{\phi}(t)$ nearest to the current time offset from the closest R-peak and subtract via Eq. (8): $\hat{s}(\hat{t}_k + p/f_s) = d_k(\phi_p) - \hat{d}_r(\phi_p)$. Samples outside the template window are passed through unchanged. Because phase-asynchronous noise is zero-mean (Assumption 1), the template noise floor averages toward zero as N_r grows.

Fig. 2 shows the result of this two-pass procedure applied to the SeizeIT2 segment from Fig. 1: the template is built from

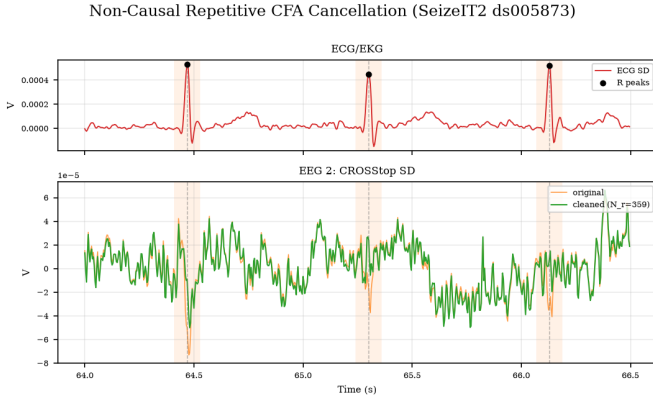


Fig. 2. Non-causal repetitive CFA cancellation on the same 3-beat segment as Fig. 1. Top: chest ECG with detected R-peaks. Bottom: original (orange) and non-causally cleaned (green) EEG 2 overlaid (MNE-preprocessed: 50 Hz notch, 0.5 Hz high-pass; template subtracted via Eq. (8)). Template estimated from the full recording session ($N_r > 300$ beats), with this 3-beat window extracted from within the cleaned signal.

the full 5-minute recording ($N_r = 359$ learnable beats), and the same 3-beat display window is extracted from the cleaned signal. Comparing the bottom panel of Fig. 2 with EEG 2 in Fig. 1, the heartbeat-locked deflections are substantially reduced, confirming that the artifact is genuinely R-peak-synchronous and addressable by the phase-indexed template.

IV. ECG PHASE ESTIMATOR

Problem 2. Estimate $\hat{t}_k$ robustly so that EEG samples can be phase-aligned via Eq. (5). The estimated cardiac phase between beats advances as

$$\hat{\phi}(t) = \frac{t - \hat{t}_k}{\hat{T}_k}, \quad \hat{T}_k = \hat{t}_{k+1} - \hat{t}_k, \quad (9)$$

where $\hat{T}_k$ is the estimated beat period.

Note 4. Since T_k is slowly varying (Assumption 3), $\hat{T}_k$ can be reliably predicted between successive R-peaks. The PI tracker advances phase continuously between beats so that $\hat{\phi}(t)$ remains well-defined in real-time causal operation.

A. Min/max tracker + PI/PID phase tracking

A QRS-emphasis band-pass feeds two streaming trackers: a peak tracker $pk(t)$ (fast attack, slow release) and a baseline tracker $bl(t)$ (slow moving average). The normalized gate

$$g(t) = \frac{x(t) - bl(t)}{pk(t) - bl(t)} \quad (10)$$

is amplitude-drift invariant, so posture/contact drift does not defeat detection. A Schmitt hysteresis and a 250 ms refractory yield clean beat gating; the fiducial t_k is the raw-reference maximum within the gated window (low jitter).

A PI phase-locked loop converts sparse fiducials into a continuous phase. Between beats the phase advances by the instantaneous rate estimate,

$$p(t+1) = [p(t) + fr(t)] \bmod 1, \quad fr(t) = \frac{1}{RR_{est}(t) f_s}, \quad (11)$$

where f_s is the sample rate. At each fiducial, a proportional term realigns phase and an integral term updates the rate estimate,

$$p \leftarrow p - K_p e_p, \quad (12)$$

$$RR_{est} \leftarrow RR_{est} + K_i (RR_{meas} - RR_{est}), \quad (13)$$

with e_p the wrapped phase error and K_p, K_i the PI gains. The loop bandwidth is set above respiratory sinus arrhythmia (RSA, ~ 0.2 – 0.4 Hz) and below per-beat detection noise. An arrhythmia supervisor classifies each beat by the ratio RR_{meas}/RR_{est} and applies the matching reflex—*premature*: trust the measured phase, freeze the rate and learning; *missed*: trust the prediction, keep the rate—protecting both the loop and the template.

B. Algorithm summary

Per sample: (1) update trackers (10) and gate; (2) advance phase (11), correct at fiducials (12)–(13), run the supervisor; (3) ILC lookup, limited subtraction, and template update on normal beats. The proposed online path is therefore constant-time per sample, with template memory linear in the window length.

V. COMPLEXITY ANALYSIS

Let L_w be the template length and C the channel count. The proposed phase-tracker + ILC path is lightweight: tracker updates and gating are constant-time, PI/PID phase update is constant-time (amortized), and ILC lookup/update is constant-time with memory L_w . Thus, per-sample complexity is $O(1)$ and memory is $O(L_w)$ for each channel. This profile is suitable for always-on low-power ear-EEG devices. For fairness, baseline adaptive cancellers (e.g., LMS/RLS) are reported separately in experiments and are not part of the proposed online deployment path.

VI. EXPERIMENTS

A. Real ear-EEG validation: coherence and repeatable-template evidence

To test the coherence assumption behind reference-driven cancellers and confirm that a genuine repeatable cardiac template exists outside simulation, we analyze a 10-minute segment (717 beats; 2 premature, 0 missed) from the SeizeIT2 dataset [12], which pairs a wearable behind-the-ear EEG channel (256 Hz) with a synchronized chest ECG lead during video-EEG epilepsy monitoring.

Coherence is near zero, even under best-case R-locked pooling. Fig. 3(a) shows magnitude-squared coherence between the ECG reference and the ear-EEG channel from long, many-average Welch segments across 0–60 Hz: coherence averages 6×10^{-10} over 8–20 Hz and is visually indistinguishable from zero across the whole spectrum – inconsistent with any stationary linear channel between $r(t)$ and $d(t)$. A skeptic could object that this washes out a relationship holding only locally around each beat; pooling short-time estimates over rolling 12-beat R-aligned batches (715 estimates, Fig. 3(b)) gives the same answer – coherence stays low and stable

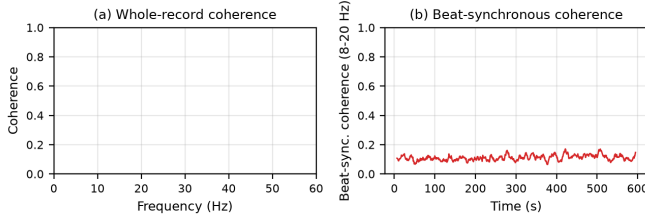


Fig. 3. Real ear-EEG/ECG coherence. (a) Whole-record Welch estimate (0–60 Hz): indistinguishable from zero across the spectrum. (b) Beat-synchronous estimate (8–20 Hz band), pooled over rolling 12-beat batches across a 10-minute recording (715 estimates): coherence stays low and stable even under best-case R-locked alignment.

(0.06–0.17, mean 0.11) throughout the recording, an order of magnitude below what a usable Hammerstein/RLS reference-driven canceller would need. Even granting the reference every possible timing advantage, the ear-EEG artifact waveform is not linearly predictable from the ECG waveform.

A repeatable, R-anchored template nonetheless exists. Coherence measures waveform predictability from the reference, not repeatability across beats. Non-causal ILC/synchronous averaging (§IV-B) applied directly to the real ear-EEG channel (Fig. 4) recovers a clear, consistent deflection time-locked to the ECG R-peak, confirming the cardiac field artifact at the ear is genuinely timing-repeatable – exactly the property ILC exploits – even though it fails the coherence test above. This is the real-data analogue of synchronous-average disk-servo RRO/NRRO separation: only the R-peak *instant* is used, never the ECG waveform shape. The causal-vs-buffered R-spike tradeoff is likewise real, not a simulation artifact: buffered matches non-causal (R-spike peak-to-peak 5.02×10^{-5} V vs. 5.07×10^{-5} V), while the causal template is capped by RR-prediction error to 3.47×10^{-5} V, a 32% reduction, consistent with §IV-B. Cardiac-band (1–20 Hz) power is reduced by 0.9/0.8/0.4 dB for non-causal/buffered/causal respectively; the modest magnitude versus the semi-synthetic ceiling (Table I) reflects broadband physiological noise at this electrode and that only the beat-locked component is addressable by ILC by construction.

These observations are the empirical basis for this paper’s scope: a null coherence result that rules out a reference-driven residual stage as a primary contribution here, and a positive repeatability result that justifies keeping the timing-driven ILC stage. The linear RLS canceller in Table I is retained only as a semi-synthetic baseline, evaluated on data explicitly given a known Hammerstein/volume-conduction channel between $r(t)$ and $d(t)$; it is not part of the proposed real-time deployment path.

B. Semi-synthetic data

The real-data evidence above has no ground truth $s(t)$, so we complement it with a semi-synthetic protocol that provides exact-reference SNR metrics; the ILC-driven design validated qualitatively above is here validated quantitatively. Following the semi-synthetic protocol common in EEG denoising [7],

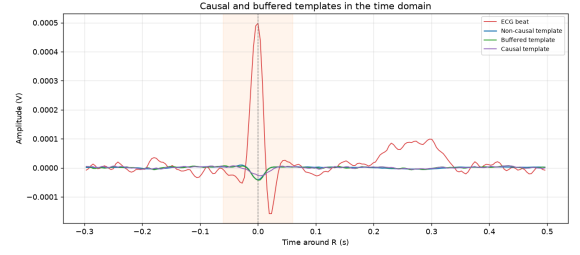


Fig. 4. Non-causal, buffered, and strict-causal templates learned from real ear-EEG (with one ECG beat shown only for R-peak timing reference). A clear, repeatable R-locked deflection is recovered despite the near-zero coherence of Fig. 3. Buffered matches non-causal; the causal (zero-latency) template’s R-spike is measurably damped by RR-prediction error, confirming the latency/suppression tradeoff of §IV-B on physiological data.

a low-CFA neural surrogate (pink noise plus an amplitude-modulated 10 Hz alpha rhythm) provides ground truth $s(t)$. A synthetic ECG source $u(t)$ with RSA-modulated RR intervals is converted into a beat-locked artifact template with slow amplitude and morphology drift, then added at a controlled input SNR; one premature and one dropped beat are injected. This yields exact-reference metrics without manual labeling. Metrics: artifact SNR (steady state), spectral cardiac-peak reduction, and alpha preservation.

C. Baselines

Within the semi-synthetic protocol only (§VI-A shows why a reference-driven residual stage cannot contribute on the real recording), we compare against (i) a linear RLS/LMS ECG-reference canceller [3]–[5]; (ii) an EKF/state-space canceller [placeholder—to be added] [10]; and (iii) low-channel ICA [placeholder—to be added] [6], which is expected to degrade at $C < 8$.

D. Results

Table I reports steady-state artifact SNR for the main approaches, generated directly by the companion script to keep manuscript numbers synchronized with code. Fig. 5 shows single-pass online convergence for the three operating modes and the corresponding buffered-mode spectrum before and after cleaning. In single-pass online operation, buffered mode converges faster and to a higher final SNR than causal mode, consistent with the expected one-beat anchoring advantage. Spectrally, the cardiac comb is strongly reduced after cleaning while the 10 Hz alpha peak is preserved within about 10%, an implicit heartbeat-evoked-potential safeguard [11]. Ablations show that phase anchoring quality and mode selection (non-causal vs buffered vs strict-causal) dominate performance: buffered ILC is closest to non-causal performance, while strict-causal ILC is limited by RR prediction accuracy around sharp R-wave transitions.

Fig. 6 shows the time-domain cancellation over three consecutive beats in the converged regime. The synthetic reference $r(t)$ exhibits full PQRS morphology—P wave, QRS complex, and T wave—rather than a simplified sinusoidal

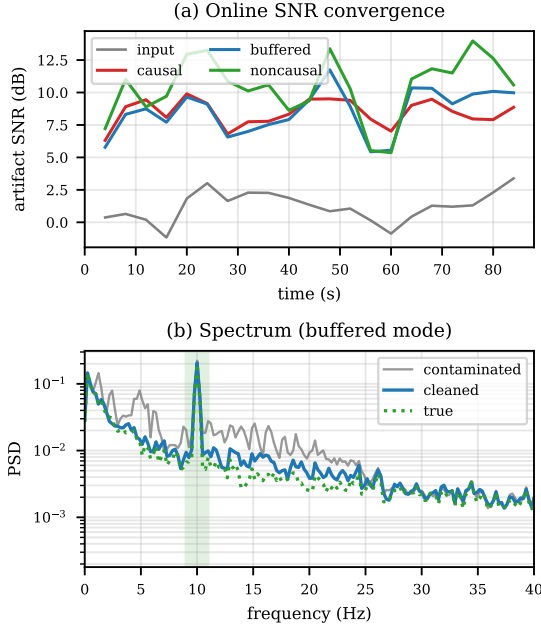


Fig. 5. (a) Online SNR convergence for the three modes (single causal pass). (b) Power spectrum before/after cleaning (buffered); shaded band marks the preserved 10 Hz alpha rhythm.

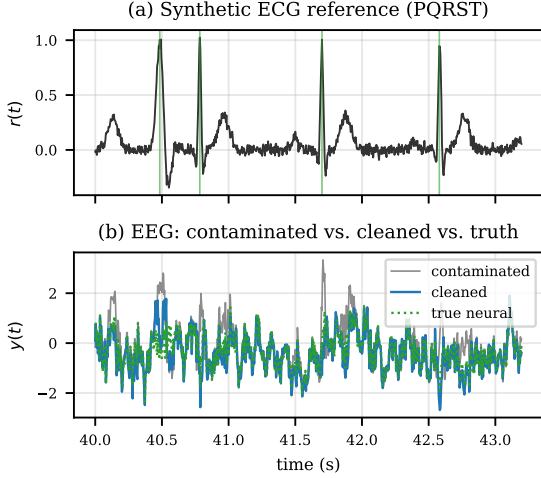


Fig. 6. Time-domain cancellation over three beats (converged, buffered mode). (a) Synthetic ECG reference $r(t)$ with full PQRST morphology; vertical lines mark detected R fiducials t_k . (b) Contaminated EEG $y(t)$, cleaned output, and ground-truth neural signal $s(t)$ over the same window.

surrogate, so the canceller must contend with a sharp, high-amplitude R deflection and a low-amplitude pre-R P wave. After processing, the cleaned EEG closely tracks the ground-truth neural signal $s(t)$: the R spike is suppressed by roughly an order of magnitude, the QRST tail is removed, and the residual P wave is attenuated, while the ongoing alpha oscillation is retained. This confirms that the phase-anchored template and residual canceller act on realistic cardiac morphology, not a reduced model.

TABLE I
STEADY-STATE ARTIFACT SNR (SEMI-SYNTHETIC; GENERATED).

Method	Latency	SNR (dB)
Input (contaminated)	–	1.27
Linear RLS ANC	causal	7.62
Proposed (causal)	0	8.58
Proposed (buffered)	1 beat	9.02
Proposed (non-causal)	offline	9.95

VII. CONCLUSION

We presented a control-theoretic synthesis—min/max-tracker PI-PLL phase estimation, ILC/repetitive template cancellation, and explicit non-causal/buffered/strict-causal operating modes—for real-time CFA removal in low-channel ear-EEG. On a real behind-the-ear EEG/chest-ECG recording, heartbeat-locked coherence between the two is near zero even under best-case R-locked pooling, while R-anchored synchronous averaging still recovers a clear, repeatable template and reproduces the causal-vs-buffered R-spike tradeoff quantitatively – evidence that, at least on this data, timing-only ILC is both necessary (reference-driven waveform regression has nothing to regress against) and sufficient (the artifact is genuinely timing-repeatable). On the semi-synthetic protocol, the method exceeds linear ANC at equal latency, exposes a quantified one-beat-latency vs. zero-latency trade-off set by RR predictability, preserves the alpha rhythm, and runs at constant per-sample cost suitable for edge deployment.

More broadly, this work aims to bridge mature mechatronic disturbance-rejection theory and biosignal denoising, opening a control-systems avenue for wearable neural interfaces. The disturbance-observer perspective extends naturally beyond the cardiac artifact: an inertial (gyroscope/accelerometer) reference could drive feed-forward cancellation of motion artifact, and an event-driven detector could trigger transient cancellation of eye-blink or jaw-clench artifacts, suggesting a unified, reference-driven, control-based artifact-rejection subsystem for earable EEG. Future work includes validation on recorded ear-EEG, multichannel spatial coupling for $C \leq 8$, an RSA-aware R predictor to lift the zero-latency R-spike ceiling, and an algorithm-unrolled low-power realization.

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